

Project 2: Mathematical Models at the Population Level to Support Immunotherapy Design

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April 29, 2026

1 Introduction

The immune system plays a crucial role in defending the body against diseases, including cancer, by identifying and eliminating abnormal cells. One of its most effective components are T cells, which recognize and attack cancerous cells. However, under conditions of chronic stimulation, such as persistent exposure to tumor-associated antigens, T cells can enter a dysfunctional phase called immune cell exhaustion.

Exhausted T cells are immune cells that have become too weak to fully eliminate the threat. This may result in a localized tumor-immune stalemate, where the tumor is contained but not completely erased. Furthermore, exhaustion is not a uniform binary state: T cells exist along a continuous spectrum ranging from fully functional to deeply exhausted. Understanding this heterogeneity is really important because even highly exhausted cells retain some functional baseline, which means that there are opportunities for immunotherapeutic interventions.

2 Immune system functioning

In a common scenario, the immune system follows a precise scheme in the fight against antigens:

Naïve T cells that find the antigen begin their first stage of reproduction and interpret the incoming signals from the environment in order to differentiate into effector cells. We can distinguish two types of effector cells, i.e. CD4+ Helper T cells and CD8+ Cytotoxic T cell (CTLs): the first ones have a coordinating role by sending chemical messages in the form of cytokines, while CTLs are the one who actually kill the infection. Once the threat has been neutralized, most of the effector cells start the apoptosis process by chemical or by Regulatory T cells. The remaining cells will differentiate into Memory T cells that can reactivate in case of a second infection of the same pathogen.

During chronic infections, such as cancer, this scheme may fail: in this case, there is an overactivation of T cells and the healthy effector cells start to become exhausted: their efficiency against the tumor starts to drastically reduce and their death rate increases, leading to a localized stalemate against the infection. In this scenario, the tumor stays at a consistent size until, because of his own nature, it develops and mutate in order to avoid the immunosuppression and start growing again.

3 Formulation of the Model

The foundational model developed by Simmons and Levy [?] describes the development of cellular exhaustion and the resulting tumor-immune stalemate. The system of ordinary differential equations (ODEs) is formulated as follows:

$$\begin{cases} \dot{N}(t) &= \alpha N \left(1 - \frac{N}{T_c}\right) - h \left(\frac{\lambda_K NK + \lambda_E NE}{N + K + E + 1}\right) \\ \dot{K}(t) &= hPK - \delta_K K \\ \dot{B}(t) &= 2^m s_B N + hPB - r_E B - \delta_B B \\ \dot{E}(t) &= r_E B - \delta_E E \\ \dot{P}(t) &= \rho_K K + \rho_B B - hP(K + B) - \delta_P P \end{cases} \quad (1)$$

Here, the state variables represent four cellular population and one cytokine respectively:

- N : Tumor volume (mm^2), modeled with logistic growth up to a carrying capacity T_c .
- K : Residual healthy CTL (CD8+) population. These cells kill tumor cells at a rate λ_K but decay at a rate δ_K .
- B : Progenitor Exhausted T cells. These are supplied through a developmental program dependent on tumor size ($2^m s_B N$) and proliferate via IL-2 (P).
- E : Terminally Exhausted T cells. They differentiate from B at a rate r_E and kill tumor cells at a reduced rate λ_E .
- P : Interleukin-2 (IL-2) cytokine concentration, produced by healthy and progenitor cells and consumed during proliferation.

Tumor killing is modeled using a saturation function to represent the physical interactions and competition in the tumor microenvironment, while the probability of contact between different populations is modeled as a product term.

4 Numerical simulations

4.1 Data reproduction

Using the parameters provided in the original study [?], the model was simulated numerically. As shown in the figures below, the system initially sees tumor growth, which drives the expansion of Progenitor (B) and Terminally Exhausted (E) cells. The healthy CTLs (K) decay over time due to low IL-2 levels. A localized tumor-immune stalemate is reached where the tumor burden stabilizes at a non-zero value, showing a chronic immune failure to completely erase the tumor.

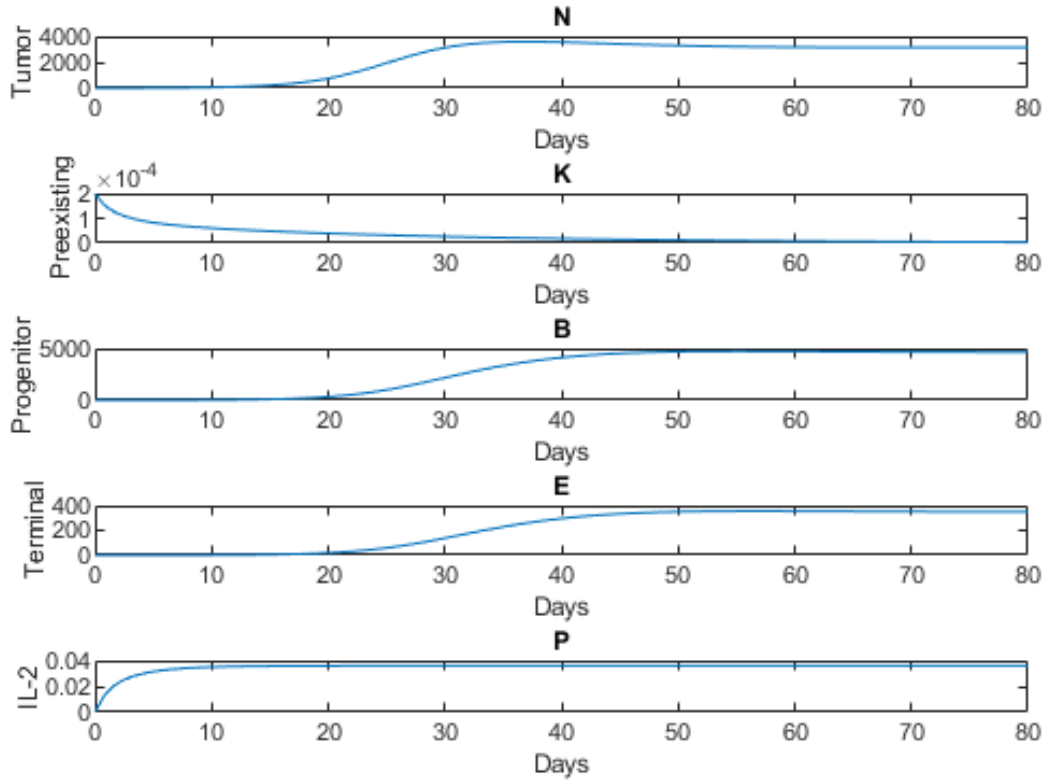


Figure 1: Reproduction of the homogeneous model dynamics showing the evolution of N , K , B , E , and P over time.

4.2 Extending the model: multiple exhaustion states

To capture the heterogeneity of T cell exhaustion, the E group of cells has been divided following different levels of exhaustion:

- E_0 : Initially Exhausted T cells (functional, capable of tumor killing).
- $E_{1/2}$: Middle-exhausted T cells (partially dysfunctional).
- E_1 : Deeply-exhausted T cells (fully exhausted, negligible killing capabilities).

The new model proposed by [?] is the following:

$$\left\{ \begin{array}{l} \dot{N}(t) = \alpha N \left(1 - \frac{N}{T_C}\right) - h \left(\frac{\lambda_k NK + \lambda_0 NE_0 + \lambda_{1/2} NE_{1/2} + \lambda_1 NE_1}{N + K + E_0 + E_{1/2} + E_1} \right), \\ \dot{K} = hPK - \delta_k K, \\ \dot{B} = 2^m S_B N + hPB - r_E B - \delta_B B, \\ \dot{E}_0 = r_E B - r_t E_0 - \delta_0 E_0, \\ \dot{E}_{1/2} = r_t E_0 - r_t E_{1/2} - \delta_{1/2} E_{1/2}, \\ \dot{E}_1 = r_t E_{1/2} - \delta_1 E_1 \\ \dot{P}(t) = \rho_K K + \rho_B B - hP(K + B) - \delta_P P, \end{array} \right. \quad (2)$$

where:

- δ_0 , $\delta_{1/2}$, and δ_1 represent the death rates for the initially, middle and deeply exhausted T cells, respectively. Deeper exhaustion is linked with a shorter lifespan due to the inability to maintain survival signals, therefore, these rates follow the inequality $\delta_0 < \delta_{1/2} < \delta_1$.
- λ_0 , $\lambda_{1/2}$, and λ_1 represent the tumor-killing rates for each corresponding sub-population. Because progressive exhaustion severely affects effector function, the killing capacity decreases as cells transition keep going, yielding to $\lambda_0 > \lambda_{1/2} > \lambda_1$.
- r_t is the phenotypic transition rate, in other words, the speed at which T cells progressively pass from one state to the other in the following order: $E_0 \rightarrow E_{1/2} \rightarrow E_1$.

4.2.1 Equivalence under Homogeneous Assumptions

First, we demonstrate that if we assume uniform killing and death rates ($\lambda_0 = \lambda_{1/2} = \lambda_1 = \lambda_E$ and $\delta_0 = \delta_{1/2} = \delta_1 = \delta_E$), the extended system turns back to the original model. In fact, if we define the total exhausted population as $E = E_0 + E_{1/2} + E_1$ and substitute the respective differential equations, we can see that:

$$\begin{aligned} \dot{E} &= \dot{E}_0 + \dot{E}_{1/2} + \dot{E}_1 \\ \dot{E} &= (r_E B - r_t E_0 - \delta_E E_0) + (r_t E_0 - r_t E_{1/2} - \delta_E E_{1/2}) + (r_t E_{1/2} - \delta_E E_1) \\ \dot{E} &= r_E B - \delta_E (E_0 + E_{1/2} + E_1) \\ \dot{E} &= r_E B - \delta_E E. \end{aligned} \quad (3)$$

By canceling the intermediate transition terms ($r_t E_0$ and $r_t E_{1/2}$), the final result matches exactly the equation for $\dot{E}(t)$ in the original system. Furthermore, if we consider also the

equation for $\dot{N}$, we got:

$$\begin{aligned}
\dot{N} &= \alpha N \left(1 - \frac{N}{T_C} \right) - h \left(\frac{\lambda_k NK + \lambda_0 NE_0 + \lambda_{1/2} NE_{1/2} + \lambda_1 NE_1}{N + K + E_0 + E_{1/2} + E_1 + 1} \right) \\
&= \alpha N \left(1 - \frac{N}{T_C} \right) - h \left(\frac{\lambda_k NK + \lambda_E NE_0 + \lambda_E NE_{1/2} + \lambda_E NE_1}{N + K + E_0 + E_{1/2} + E_1 + 1} \right) \\
&= \alpha N \left(1 - \frac{N}{T_C} \right) - h \left(\frac{\lambda_k NK + \lambda_E N(E_0 + E_{1/2} + E_1)}{N + K + (E_0 + E_{1/2} + E_1) + 1} \right) \\
&= \alpha N \left(1 - \frac{N}{T_C} \right) - h \left(\frac{\lambda_k NK + \lambda_E NE}{N + K + E + 1} \right)
\end{aligned} \tag{4}$$

confirming that, with homogeneous assumptions, the model obtained is again the original one. This equivalence is validated numerically below.

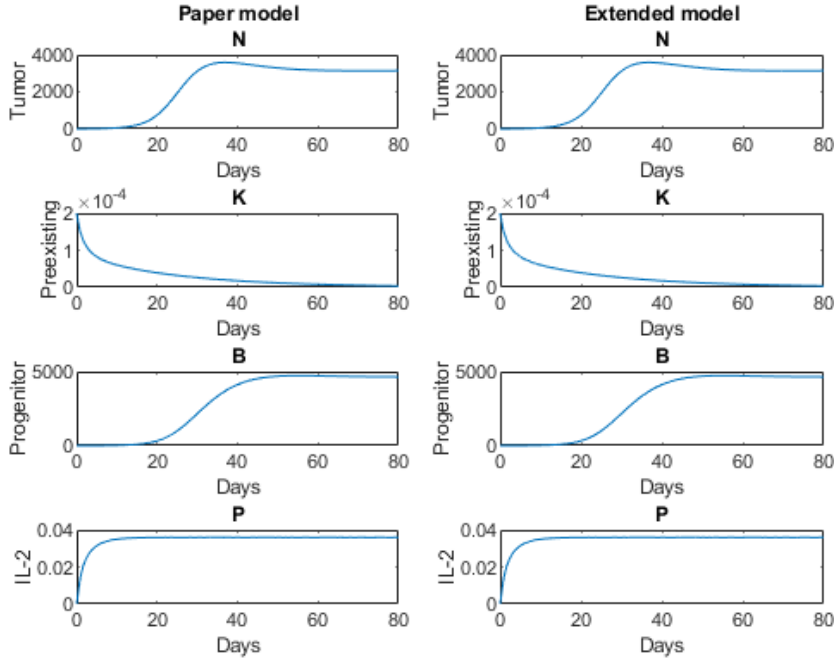


Figure 2: Comparison between the original system and the new one with $r_t = 5r_E$

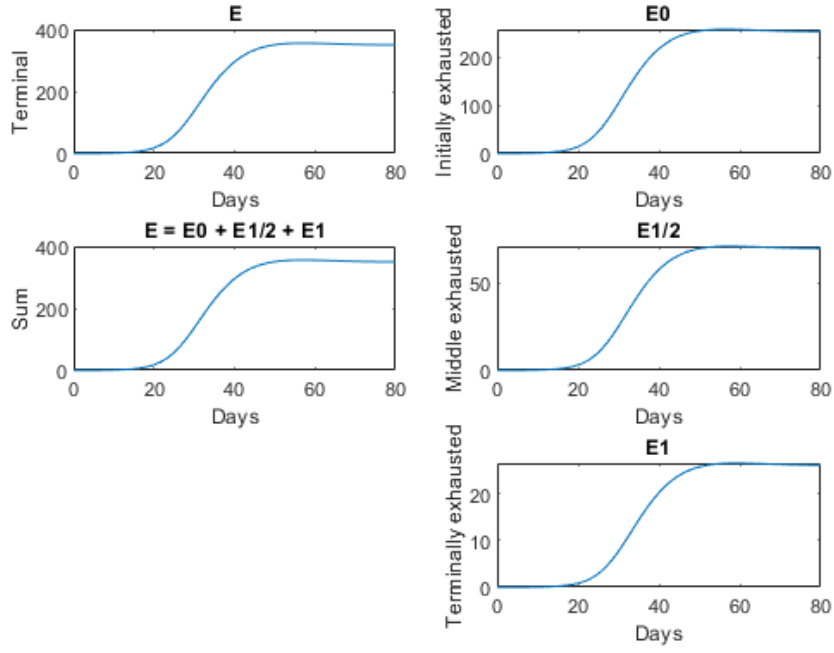


Figure 3: Comparison with $r_t = 5r_E$ showing that the sum of the extended compartments ($E_0 + E_{1/2} + E_1$) perfectly overlaps with the E compartment from the homogeneous paper model.

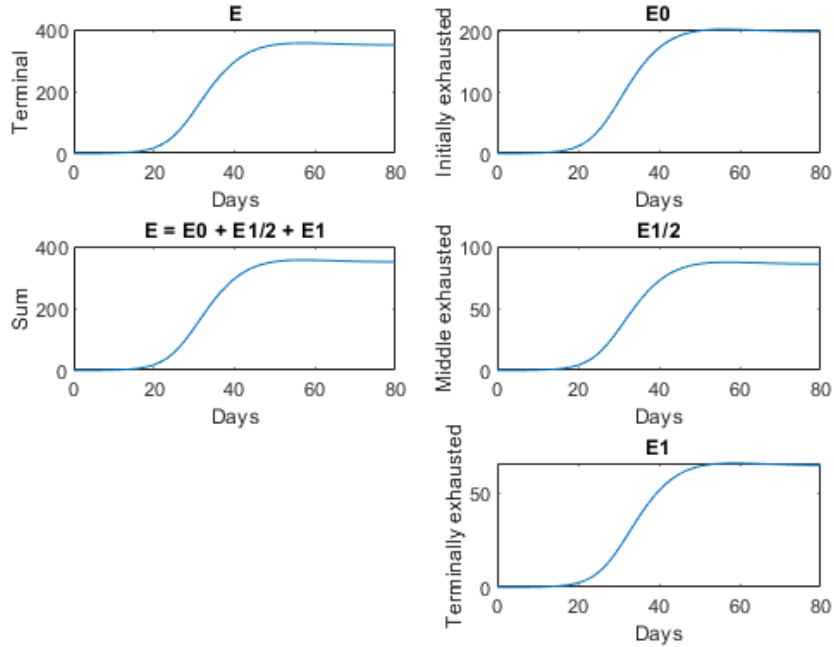


Figure 4: Comparison with $r_t = 10r_E$ showing that the sum of the extended compartments ($E_0 + E_{1/2} + E_1$) perfectly overlaps with the E compartment from the homogeneous paper model.

It is good to notice that while the sum $E_0 + E_{1/2} + E_1$ remains the the same with both

values of r_t , the composition of the sum is different:

- When r_t is lower ($r_t = 5r_E$, Figure 3), the transition is slower, causing most cells to remain in an initial exhaustion state (E_0).
- When r_t is doubled ($r_t = 10r_E$, Figure 4), cells become exhausted much faster. As a result, the peak E_0 population drops, while the deeply exhausted groups ($E_{1/2}$ and E_1) grow more.

4.2.2 Introduction of heterogeneity in killing and death rates

Deeply exhausted T cells have shorter lifespans and lower tumor-killing capacities. To simulate that, we define a continuous linear function for the killing rate $\lambda(y)$ and death rate $\delta(y)$ as functions of $y \in [0, 1]$:

$$\lambda(y) = \lambda_K + y(\lambda_E - \lambda_K)$$

$$\delta(y) = \delta_K + y(\delta_E - \delta_K)$$

Using these linear functions, we evaluate the intermediate state ($y = 1/2$) to derive $\lambda_{1/2}$ and $\delta_{1/2}$:

$$\lambda_{1/2} = \lambda_K + \frac{1}{2}(\lambda_E - \lambda_K)$$

$$\delta_{1/2} = \delta_K + \frac{1}{2}(\delta_E - \delta_K)$$

With this choice, we assumed that Initially Exhausted T cells ($y = 0$) have the same killing/death rate as Healthy T cells and that Deeply Exhausted T cells ($y = 1$) have the same killing/death rate as the Terminally T cells.

4.3 Simulation and Analysis of the Extended Model

The implementation of these heterogeneous rates modify the dynamics of the tumor-immune interaction, as shown below.

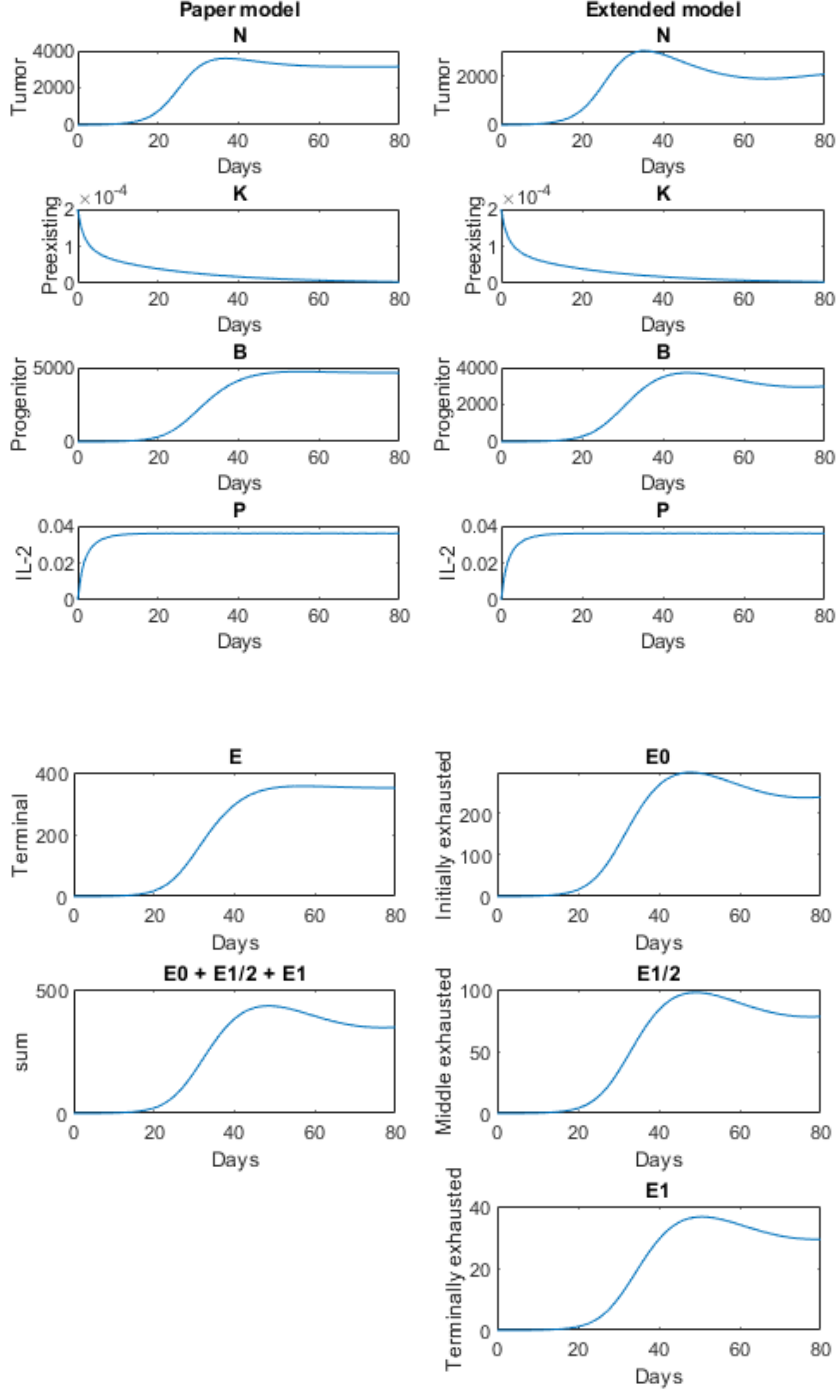


Figure 5: Evolution of variables under heterogeneous assumptions compared to the original system with $r_t = 5 \cdot r_E$.

By adding specific killing and death rates based on the exhaustion phenotype, the initially exhausted cells E_0 provide a higher immediate killing capacity, comparable to fully functional CTLs. However, their transition into deep exhaustion E_1 together with higher death rates changes the size and stability of the tumor-immune stalemate. Specifically, the strong early response from E_0 drives a sharper decline in tumor volume, establishing a new equilibrium at a significantly lower tumor burden than the one predicted by the homogeneous model. This creates a huge domino effect on the other populations: since the

presence of Progenitor cells (B) is strictly dependent on tumor size through the developmental program term $2^m s_B N$, this smaller tumor burden provides less antigen stimulation. As a result, the Progenitor population (B) peaks and then declines to a lower steady-state compared to the homogeneous model. This smaller supply restricts the growth of E_0 . As a result, fewer cells move on to the later $E_{1/2}$ and E_1 stages. This explains also why the total exhausted sum ($E_0 + E_{1/2} + E_1$) settles at a lower equilibrium after its initial, highly effective spike. In conclusion, having higher functional cells (E_0) clears the tumor faster, but removing the chronic antigen stimulation the immune system cuts off its own supply of T cells, blocking into a permanent stalemate.

5 Continuous Population Model

Biological phenomena involving phenotypic variation can often be more accurately modeled as continuous spectra rather than discrete compartments. In this section we will formulate an extension of the model introducing a continuous phenotype $y \in [0, 1]$. We will define $E(t, y)$ as a density function representing the total number of exhausted cells at time t characterized by phenotype y , governed by a Partial Differential Equation.

5.1 Derivation of the model

Starting from the original model equation

$$\frac{dE_{tot}(t)}{dt} = r_E B - \delta_E E_{tot} \quad (5)$$

we want to distinguish cells based on their exhaustion level. We define a density function $E(t, y)$ such that the total number of cells is expressed by

$$E_{tot}(t) = \int_0^1 E(t, y) dy. \quad (6)$$

If we consider a small piece of the space Δy at time t , we have that the number of cells in this piece is defined by:

- number of still functional cells,
- number of cells that mutate towards exhaustion
- number of dying cells.

Introducing $v(t, y)$ as the cells aging speed, the equation becomes:

$$\frac{\partial}{\partial t} [E(t, y) \Delta y] = v(t, y) E(t, y) - v(t, y + \Delta y) E(t, y + \Delta y) - \delta_E E(t, y) \Delta y \quad (7)$$

and dividing by Δy , we obtain the incremental ratio

$$\frac{\partial}{\partial t} E(t, y) = \frac{v(t, y) E(t, y) - v(t, y + \Delta y) E(t, y + \Delta y)}{\Delta y} - \delta_E E(t, y). \quad (8)$$

Taking the limit $\Delta y \rightarrow 0$, we obtain

$$\frac{\partial}{\partial t}E(t, y) = -\frac{\partial}{\partial y}(v(t, y)E(t, y)) - \delta_E E(t, y) \quad (9)$$

rearranging the terms, we have an advection-reaction PDE:

$$\frac{\partial}{\partial t}E(t, y) + \frac{\partial}{\partial y}(v(t, y)E(t, y)) = -\delta_E E(t, y). \quad (10)$$

To have a unique solution we have to put some boundary conditions, and we choose them in order to respect the initial equation:

- At the beginning of exhaustion, we know that the exhausted cells come from the term $r_t B(t)$, that we can consider as the incoming flux on the left side, i.e.

$$v(t, 0)E(t, 0) = r_E B(t) \quad (11)$$

- To stop the population at the final state ($y = 1$) we have to impose that the aging speed is 0, i.e.

$$v(t, 1) = 0. \quad (12)$$

Once we have a well defined PDE, we are left with adjusting the parameters λ_E and δ_E in order to change in function of the phenotype y :

$$\begin{aligned} \lambda_t(y) &= \lambda_K + y(\lambda_E - \lambda_K) \\ \delta_t(y) &= \delta_K + y(\delta_E - \delta_K) \end{aligned} \quad (13)$$

so the final system of ODEs and PDEs is the following:

$$\left\{ \begin{aligned} \dot{N}(t) &= \alpha N \left(1 - \frac{N}{T_c}\right) - h \left(\frac{\lambda_K N K + N \int_0^1 \lambda_t(y) E(t, y) dy}{N + K + E_{tot} + 1} \right) \\ \dot{K}(t) &= h P K - \delta_K K \\ \dot{B}(t) &= 2^m s_B N + h P B - r_E B - \delta_B B \\ \dot{P}(t) &= \rho_K K + \rho_B B - h P (K + B) - \delta_P P \\ \frac{\partial E(t, y)}{\partial t} &= -\frac{d}{dy} v(t, y) E(t, y) - \delta_t(y) E(t, y) \\ v(t, 0) E(t, 0) &= r_E B(t) \\ v(t, 1) &= 0. \end{aligned} \right. \quad (14)$$

5.2 Numerical approach

$$J = \begin{bmatrix} A_{ODE} & C_{int} & B_{inp} & A_{PDE} \end{bmatrix}$$